MILITARY SPECIFICATION

2,4 PENTANEDIONE

This Interim Amendment is approved for use within the US Army Armament, Munitions and Chemical Command, with MIL-P-47067, dated 10 May 1974.

PAGE 1

1. Scope. Change "one grade" to "two grades".

PAGE 2

2.2 Other Publications. Add the following to the end of the paragraph:

"AMERICAN SOCIETY FOR TESTING AND MATERIALS

ASTM E203 - Water Using Karl Fischer Reagent
ASTM E300 - Recommended Practice for Sampling Industrial Chemicals

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103).

CODE OF FEDERAL REGULATIONS

Title 49, Transportation, Parts 100-199

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, D.C. 20402. Orders for the above publication should cite: 49 CFR 100-199 (latest revision).

UNITED NATIONS

United Nations - Transport of Dangerous Goods (Recommendations of the Committee of Experts on the Transport of Dangerous Goods)."

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3.1 Properties. Add the following paragraph:

"3.1.1 Properties. (for M864 Propellant Component only). The material shall conform to the following requirements:

Property	Min	Max	Applicable Test Method
Assay, %	98	-	4.4.1.1
Moisture, %	-	0.2	4.4.3
Refractive Index	1.4505	1.4525	4.4.4
Specific Gravity, 20°C	0.9722	0.978	4.4.2"

PAGE 3

4.3 Sampling. Add the following paragraph:

"4.3.1 Sampling for M864 Propellant Component. Approximately 500 grams of the composition shall be selected from each batch to be sampled using ASTM Method E300-73 for solids. Samples shall be selected for inspection in accordance with table below. If any sample fails to meet any test requirement the batch represented by the sample shall be rejected. Each required test shall be performed in duplicate. No composite samples shall be used.

Number of containers in batch	Number of containers to be sampled
2 - 100	2
100 - 200	5
200 - 300	8
300 - 500	10
500 - 1000	13
1000 - 3000	20
3000 - 10,000	32"

PAGE 6

4.4.1 Assay. Add the following new paragraphs:

"4.4.1.1 Assay. (for M864 propellant component only). The Assay of the material shall be determined as follows:

a. Appartus
Instrument: Varian 2700 or equivalent with Flame Ionization Detector.
Column: 6' x 1/8" stainless steel, packed with 10% SP-1000 on suplecoport.

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b. Instrument Settings
Column temperature: Programmed from 50 to 220°C at 8.0°C/minute.
Detector temperature: 310°C
Carrier Gas: Helium at 40 ml/min
Sample Size: 0.2

c. Calculation. Measure the area under all peaks by any convenient method and calculate the product concentration in area percent.

4.4.2 Specific Gravity. Add the following new paragraph:

"4.4.2.1 Specific Gravity. (for M864 propellant component only.)

a. Apparatus. Side arm pycnometer, with thermometer and ground glass joint and cap for side-arm, also with ground glass joint.

b. Procedure.

1. Assemble and weigh the empty pycnometer to the nearest 0.1mg.

2. Fill the pycnometer with distilled water to the middle of the ground glass joint.

3. If measurement of specific gravity (and/or density) is to be made above room temperature, wrap in tissue and warm with hand. If liquid is to be measured below room temperature, cool water to a few degrees below desired temperature.

4. Carefully insert the thermometer so that no air is trapped; wipe pycnometer dry.

5. When exact temperature is reached, remove any surplus liquid from the end of the side-arm and place the cap in position. (Liquid must fill side-arm at the specified temperature).

6. Carefully weigh to the nearest 0.1 mg.

7. Empty and carefully dry the pycnometer.

8. Fill the pycnometer with liquid to be measured and repeat steps 3-7.

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c. Calculations.

1. Specific gravity = B ÷ A

Where A = mass of water contained in pycnometer (g)

B = mass of liquid contained in pycnometer (g)

2. Volume of Pycnometer.

Volume = $(W_2 - W_1) \div d$

Where: W_2 = weight of pycnometer plus water.

W_1 = weight of empty, dry pycnometer.

d = density of water at t^o, g/ml (obtain from CRC handbook)

t^o = temperature of calibration

3. Density.

$$\text{Density, g/ml} = \frac{W_3 - W_1}{\text{volume of pycnometer}}$$

Where: W_3 = weight of sample plus pycnometer (grams).

W_1 = weight of empty, dry pycnometer (grams)."

Add the following new paragraphs:

"4.4.3 Moisture. (For M864 propellant component only). The moisture content shall be determined in accordance with ASTM Method E203.

4.4.4 Refractive Index. (for M864 Propellant component only).

4.4.4.1 Apparatus.

a. Abbe-3L Refractometer (Bausch and Lomb) or equivalent

b. Solvent for sample

c. Soft tissues

d. Acetone

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4.4.4.2 Procedure.

a. Clean the upper and lower prisms with soft, nonabrasive, absorbent tissues wetted with acetone; allow them to dry.

b. Introduce the sample upon the prism by means of a pipette or dropper. The prisms are then closed slowly allowing the excess to squeeze out into the space between the metal mounts. NEVER USE GLASS OR METAL APPLICATORS AGAINST THE PRISM FACE.

c. After the sample is in position on the instrument, set the scale at the approximate value expected. (To see the scale, depress the momentary contact switch on the side of the instrument.)

d. Release the switch. Bring the borderline which will probably be strongly colored near the crosshair and compensate the color by adjusting the position of the dial. The borderline should be faintly blue on one side and faintly red at the other.

e. Observe the crosshair, sharply focusing the eyepiece if necessary and bring the dividing line upon their intersection by means of the coarse or fine hand controls.

f. Read the index by depressing themomentary contact switch. Estimate the fourth place.

4.4.4.3 Calculations.

a. Refractive Index is read directly from the scale.

b. The refractive index decreases with a temperature rise because the density decreases, resulting in fewer molecules per unit volume. The prisms can be temperature-controlled to meet the need. Free-flowing liquids can be introduced with an eyedropper into the channel which is alongside the prisms. When a mixture to be analyzed contains a volatile component, this technique is particularly useful because it minimizes the loss by evaporation of the volatile component and this affects the accuracy of the readout. Never touch the prisms with any hard object. Always clean with soft tissues moistened with alcohol, acetone, or ether; the prisms after being cleaned, should always be left in a locked position so that dirt and dust will not collect on them."

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PAGE 7

Delete Section 5 in its entirety and substitute with the following:

"5. Packaging.

5.1 Packaging and Packing.

5.1.1 Level A. - When applicable packaging shall be as specified in the contract.

5.1.2 Level B. - Unless otherwise specified in the contract, packaging and packing shall be in accordance with the standard commercial practice applicable to the type of material. The packaging and packing shall be of such construction and materials that the contents will be adequately protected against damage, loss, or contamination when shipped from the contractor or supply source to the receiving activity, or stored for one year from the date of manufacture. Container size/capacity shall be as specified in the contract. The type of containers and packing materials selected shall comply with applicable carrier rules and regulations. In addition, any hazardous material must be packed and shipped in accordance with the Code of Federal Regulations (CFR), Title 49, Parts 100-199, and also when offered for overseas shipment, the United Nations-Transport of Dangerous Goods (Recommendations of the Committee of Experts on the Transport of Dangerous Goods). It shall be the responsibillty of the supplier to obtain all data necessary from DOT, and if applicable, from the Committee of Experts on the Transport of Dangerous Goods (UN-Transport of Dangerous Goods) to ship hazardous materials legally.

5.1.3 Level C. - Same as Level B, except not authorized for overseas shipment and that the United Nations-Transport of Dangerous Goods does not apply.

5.2 Marking. In addition to any special marking required by the contract, each container shall be marked in accordance with MIL-STD-129, the code of Federal Regulations, (CFR), Title 49, Parts 100-199, and the United Nations-Transport of Dangerous Goods (Recommendations of the Committee of Experts of the Transport of Dangerous Goods). Marking shall include but not be limited to, the following information.

a. Title, Number and Date of this Specification

b. Manufacturer's Name and Location

c. Material Trade Name

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d. Net Weight and Volume

e. Lot Number, Batch Number(s), and Date of Manufacture

f. Storage Conditions/Requirements

g. Toxicity Precautions, if applicable

h. Contract Number

i. Warranty Provisions

6.1 Intended use. Delete "motors." and substitute "motors including M864 propellant."

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Preparing activity
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(Project 6810-1063)

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